

Output

Case 0: Correct workflow

1. Admin logins and added candidates

```

      ONLINE VOTING SYSTEM
1. Admin
2. Voter
3. Exit

Enter Choice: 1
Enter Admin Password: admin123

      ADMIN MENU
1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back
Enter Choice: 1
Enter number of candidates: 2
Enter Candidate Name: a
Enter Candidate Name: b
Candidates Added Successfully!
```

2. Admin registered voters and started election

```
ADMIN MENU
1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back
Enter Choice: 2
Enter number of voters: 3
Enter Voter ID: 1
Enter Voter Name: q
Set Password: 1q
Enter Voter ID: 2
Enter Voter Name: w
Set Password: 2w
Enter Voter ID: 3
Enter Voter Name: e
Set Password: 3e
Voters Added Successfully!
```

```
ADMIN MENU
1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back
Enter Choice: 3
Election Started Successfully!
```

3. Voters casted votes

ADMIN MENU

1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back

Enter Choice: 7

Returning...

ONLINE VOTING SYSTEM

1. Admin
2. Voter
3. Exit

Enter Choice: 2

Enter Voter ID: 1

Enter Password: 1q

CANDIDATES

1. a
2. b

Enter Candidate ID: 1

Vote Cast Successfully!

ONLINE VOTING SYSTEM

1. Admin
2. Voter
3. Exit

Enter Choice: 2

Enter Voter ID: 2

Enter Password: 2w

CANDIDATES

1. a
2. b

Enter Candidate ID: 2

Vote Cast Successfully!

ONLINE VOTING SYSTEM

1. Admin
2. Voter
3. Exit

Enter Choice: 2

Enter Voter ID: 3

Enter Password: 3e

CANDIDATES

1. a
2. b

Enter Candidate ID: 1

Vote Cast Successfully!

4. Admin displayed role

```
ONLINE VOTING SYSTEM
1. Admin
2. Voter
3. Exit

Enter Choice: 1
Enter Admin Password: admin123

ADMIN MENU
1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back
Enter Choice: 6
Election Officer is an Admin
```

5. Admin ended election and viewed results

```
ADMIN MENU
1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back
Enter Choice: 4
Election Ended Successfully!

ADMIN MENU
1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back
Enter Choice: 5

FINAL RESULTS
1. a : 2 votes
2. b : 1 votes

Winner is: a with 2 votes
```

6. Exiting

```
ADMIN MENU
1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back
Enter Choice: 7
Returning...

ONLINE VOTING SYSTEM
1. Admin
2. Voter
3. Exit

Enter Choice: 3
Exiting System...
```

Exception Cases:

Case 1: Trying to vote before admin registers voters

```
ONLINE VOTING SYSTEM
1. Admin
2. Voter
3. Exit

Enter Choice: 2
Exception: No voters registered!
```

Case 2: Admin trying to start election before entering candidates or voters list

<pre>ONLINE VOTING SYSTEM 1. Admin 2. Voter 3. Exit Enter Choice: 1 Enter Admin Password: admin123 ADMIN MENU 1. Add Candidates 2. Add Voters 3. Start Election 4. End Election 5. View Results 6. Display Role 7. Back Enter Choice: 3 Add candidates first!</pre>	<pre>ADMIN MENU 1. Add Candidates 2. Add Voters 3. Start Election 4. End Election 5. View Results 6. Display Role 7. Back Enter Choice: 1 Enter number of candidates: 2 Enter Candidate Name: a Enter Candidate Name: b Candidates Added Successfully! ADMIN MENU 1. Add Candidates 2. Add Voters 3. Start Election 4. End Election 5. View Results 6. Display Role 7. Back Enter Choice: 3 Add voters first!</pre>
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Case 3: Admin trying to view vote count before election ends

```
ADMIN MENU
1. Add Candidates
2. Add Voters
3. Start Election
4. End Election
5. View Results
6. Display Role
7. Back
Enter Choice: 5
Results can be viewed only after election ends!
```

Case 4: Incorrect Id/password entered by voter

<pre>ONLINE VOTING SYSTEM 1. Admin 2. Voter 3. Exit Enter Choice: 2 Enter Voter ID: 1 Enter Password: 2w Exception: Incorrect Password!</pre>	<pre>ONLINE VOTING SYSTEM 1. Admin 2. Voter 3. Exit Enter Choice: 2 Enter Voter ID: 3 Exception: Voter not found!</pre>
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Case 5: Incorrect password by admin

```
ONLINE VOTING SYSTEM
1. Admin
2. Voter
3. Exit

Enter Choice: 1
Enter Admin Password: incorrect123
Invalid Password!
```


Case 6: Voter trying to vote more than once

```

      ONLINE VOTING SYSTEM
1. Admin
2. Voter
3. Exit

Enter Choice: 2
Enter Voter ID: 1
Enter Password: 1q

      CANDIDATES
1. a
2. b
Enter Candidate ID: 1
Vote Cast Successfully!

      ONLINE VOTING SYSTEM
1. Admin
2. Voter
3. Exit

Enter Choice: 2
Enter Voter ID: 1
Enter Password: 1q

      CANDIDATES
1. a
2. b
Enter Candidate ID: 1
Exception: You have already voted!
```


Case 7: Incorrect candidate Id

```
      ONLINE VOTING SYSTEM
1. Admin
2. Voter
3. Exit

Enter Choice: 2
Enter Voter ID: 1
Enter Password: 1q

      CANDIDATES
1. a
2. b
Enter Candidate ID: a
Exception in thread "main" java.util.InputMismatchException
    at java.base/java.util.Scanner.throwFor(Scanner.java:977)
    at java.base/java.util.Scanner.next(Scanner.java:1632)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2297)
    at java.base/java.util.Scanner.nextInt(Scanner.java:2251)
    at OnlineVotingSystem.main(OnlineVotingSystem.java:334)
```